

TRENTOS on ODROID-C2

Page Content

- [Hardware](#)
- [Documentation](#)
- [Wiring the board](#)
- [Build Process](#)
 - [Binaries](#)
 - [Build ATF \(optional\)](#)
 - [Compile U-Boot](#)
 - [Configure PXE Boot](#)
 - [Obtain prebuilt bootloader/firmware binaries](#)
 - [Generate bootloader image](#)
- [Flashing Process](#)
 - [Create partitions](#)
 - [Flash bootloader images](#)
 - [microSD layout](#)
- [Custom Setup](#)
- [Bootflow](#)
 - [Binaries](#)
 - [Memory usage during the boot process](#)
 - [Bootup](#)
- [seL4/TRENTOS support](#)
- [UARTs](#)

Hardware

- Amlogic Meson S905 (GXBB)
 - 4x Cortex-A53 (1.5 GHz)
 - Mali-450 MP3 GPU (700 MHz, OpenGL ES 2.0/1.1)
 - Cortex-M3 as SCP
- Board
 - 2 GB DDR3-RAM
 - Gigabit MAC, Realtek RTL8211F PHY
 - 4x USB 2.0 (Genesys Logic GL852G)
 - 1x USB 2.0 OTG
- no longer available after Q1/2021

Documentation

- <https://www.hardkernel.com/shop/odroid-c2/>
- <https://wiki.odroid.com/odroid-c2/odroid-c2>

Wiring the board

The ODROID-C2 requires the following physical setup:

- a power supply (5V2A)
 - either via the DC-Jack (using an adapter with inner (positive) diameter 0.8mm and outer (negative) diameter 2.5mm)
 - or via the USB OTG port (using a microUSB cable)
 - https://wiki.odroid.com/odroid-c2/hardware/otg_power_enable
- a serial connection to a PC via a USB-to-UART adapter (3.3V) to the [4-pin CON5 connector](#)
- a microSD card or eMMC (2-3 times faster than SD card)
- in case PXE boot shall be used, an Ethernet cable has to be connected to the board's Ethernet interface, in order to receive relevant data from the PC

Build Process

Binaries

Amlogic ARMv8 based SoCs use a vendor variant of the Trusted Firmware-A boot architecture. The following table clusters the different boot loader stages (according to U-Boot [pre-generated-fip](#)):

Boot Loader Stage	Type (according to TF-A)	File	Executed on	Exception Level	Status	Delivered as	Implementations	Description
???	Boot ROM/AP Trusted ROM	-	Cortex-A53	EL3	discarded after boot	prebuilt	platform provider specific	
BL1	Boot ROM/AP Trusted ROM	bl1.bin.hardkernel	Cortex-A53	EL3	discarded after boot	prebuilt	- platform provider specific - ARM Trusted Firmware BL1	First boot step, implemented in ROM on Amlogic SoCs Question: actually a file on SD/eMMC, so not running in ROM
BL2	Trusted Boot Firmware	bl2.package	Cortex-A53	S-EL1	discarded after boot	prebuilt	- platform provider specific - ARM Trusted Firmware BL2	Second boot step, used to initialize the SoC main clocks & DDR interface. The BL21 and ACS board-specific binaries are "inserted" in the BL32 binary before signing/packaging in order to be flashed on the platform.
BL30	-	bl30.bin	Cortex-M3	-	runtime	prebuilt	platform provider specific	Amlogic Secure Co-Processor (SCP) firmware used to handle all the system management operations (DVFS, suspend/resume, ...)
BL301	-	bl301.bin	Cortex-M3	-	runtime	prebuilt	platform provider specific	Amlogic Secure Co-Processor (SCP) board-specific firmware "plug-in" to handle custom DVFS & suspend-resume parameters
BL31	EL3 Runtime Firmware (Secure Monitor)	bl31.bin	Cortex-A53	EL3	runtime	prebuilt (but available as source code)	- platform provider specific - ARM Trusted Firmware BL31	Initializes the interrupt controller and the system management interface (PSCI)
BL32 (Optional)	Secure-EL1 Payload (optional)	-	Cortex-A53	S-EL1	runtime	-	-	The Trusted Environment Execution (TEE) Operating System to run secure Trusted Apps, e.g. OP-TEE
BL33	Non-trusted Firmware	u-boot.bin	Cortex-A53	NS-EL2	runtime	available as source code	U-Boot	The last non-secure step, usually U-Boot which loads Linux

Build ATF (optional)

TODO: official ATF can be built from source, if required

- build instructions at <https://trustedfirmware-a.readthedocs.io/en/latest/plat/meson-gxbb.html>
- otherwise, a respective prebuilt image **bl31.bin** is already provided for download as part of the amlogic-boot-fip repository by LibreELEC (<https://github.com/LibreELEC/amlogic-boot-fip/tree/master/odroid-c2>)

Compile U-Boot

Mainline U-Boot source code has to be obtained from the official GitLab repository (<https://source.denx.de/u-boot/u-boot>). Afterwards, U-Boot can be built for the ODROID-C2 via the following commands (documented at <https://u-boot.readthedocs.io/en/latest/board/amlogic/odroid-c2.html>):

```
$ export CROSS_COMPILE=aarch64-linux-gnu-
$ make odroid-c2_defconfig
$ make
```

The resulting U-Boot image **u-boot.bin** can be found in the U-Boot root folder.

Configure PXE Boot

The ODROID-C2 regularly boots from microSD card or eMMC. For development purposes it is nevertheless recommended to stick to a PXE setup. For this, we assume tftphoot is already configured on the PC side. Otherwise, please see the guide available [here](#).

In order to allow for PXE boot, a respective U-Boot configuration in form of a **boot.scr** file has to be provided. Therefore, at first, an empty file **boot.script** has to be created. This file then has to be edited and filled with the following content:

```
setenv load_addr 0x40000000

setenv ipaddr 10.0.0.11
setenv serverip 10.0.0.10

tftp ${load_addr} os_image.elf
setenv autostart yes
bootelf ${load_addr}
```

Afterwards, the **boot.script** file has to be converted into a **boot.scr** file. Therefore, the **mkimage** tool is required, which is provided as part of the **u-boot-tools**. If not available on the PC, the tools can be installed via

```
$ sudo apt install u-boot-tools
```

The actual conversion is then provided in form of

```
$ mkimage -c none -A arm -T script -d boot.script boot.scr
```

Obtain prebuilt bootloader/firmware binaries

Prebuilt bootloader/firmware images for the Amlogic S905 SoC in the ODROID-C2 are provided by Amlogic. These files can be obtained from the **amlogic-boot-fip** repository provided by LibreELEC (<https://github.com/LibreELEC/amlogic-boot-fip>). The relevant parts for the ODROID-C2 are stored in the subfolder **odroid-c2** and comprise the following files:

- **bl1.bin.hardkernel**
- **bl2.package**
- **bl30.bin**
- **bl31.bin**
- **bl301.bin**

In addition, the folder also contains two tools that are required for creating and signing the final images:

- **aml_encrypt_gxb**
- **fip_create**

Generate bootloader image

At first, the **fip_create** tool has to be used in order to create a Firmware Image Package (FIP), named **fip.bin**, out of the prebuilt images **bl30.bin**, **bl301.bin** and **bl31.bin** as well as the generated **u-boot.bin**:

```
$ ./fip_create --bl30 bl30.bin --bl301 bl301.bin --bl31 bl31.bin --bl33 u-boot.bin fip.bin
```

Afterwards, we print the contents of FIP:

```
$ ./fip_create --dump fip.bin
```

The content of **fip.bin** looks like the following:

```
0x000000: FIB header
0x040000: BL30 (SCP Firmware), size=0x009E88
0x100000: BL301 (SCP Firmware), size=0x001854
0x140000: BL31 (EL3 Runtime), size=0x0110D0
0x280000: BL33 (Custom BL), size=0x0AD138
```

Then, we bring together **bl2.package** and the FIP into the file **boot_new.bin**:

```
$ cat bl2.package fip.bin > boot_new.bin
```

This leads to the following content structure:

```
0x000000: bl2.package
0x0C0000: FIB header
0x100000: BL30 (SCP Firmware), size=0x009E88
0x1C0000: BL301 (SCP Firmware), size=0x001854
0x200000: BL31 (EL3 Runtime), size=0x0110D0
0x340000: BL33 (Custom BL), size=0x0AD138
```

The resulting image now has to be signed with the help of the ***aml_encrypt_gxb*** tool and is renamed to ***u-boot.img***:

```
$ ./aml_encrypt_gxb --bootsig --input boot_new.bin --output u-boot.img
```

An alternative to the proprietary toolchain might be provided by the open source [meson-tools](#).

Finally, the content of ***u-boot.img*** is copied into the file ***u-boot.gxbb***, thereby skipping the first 96 blocks (here: $96 * 512\text{B} = 48\text{ KiB}$) during reading.

```
$ dd if=u-boot.img of=u-boot.gxbb bs=512 skip=96
```

Flashing Process

Create partitions

Afterwards, we create two separate partitions (***/boot*** and ***/rootfs***). The ***/boot*** partition shall then contain the U-Boot environment (***boot.scr***) that allows for setting up U-Boot with the PXE boot.

```
$ DEV=/dev/sda

# create two partitions (/boot and /rootfs) on the microSD card
# - partition /boot - FAT32 / 1GB
# - partition /rootfs - EXT4
# blank lines in the command correspond to pressing 'Enter' to accept default values from fdisk

$ sudo fdisk /dev/sda << EOF
n
p
1

+1G
t
b
n
p

t

83
w
EOF

# create file system for partition 1
$ sudo mkfs.vfat /dev/sda1 -n boot

# create file system for partition 2
$ sudo mkfs.ext4 /dev/sda2 -L rootfs

# copy boot.scr to /boot partition
$ sudo cp boot.scr <path/to/boot/partition>
```

Flash bootloader images

After generating the bootloader image, both contents from **u-boot.gxbb** and **bl1.bin.hardkernel** have to be installed either in microSD card or the eMMC. Exemplarily, the card is to be seen as **/dev/sda**. This identifier must be adapted according to the individual setup. The actual flashing can then be done via the following commands:

```
$ DEV=/dev/sda

# Write the image to SD. Sector 0 holds the MBR, populate the first
# 442 byte with a boot loader. The space from byte 443 to the end is
# left untouched, because this contains the partition information
$ sudo dd if=bl1.bin.hardkernel of=$DEV conv=fsync bs=1 count=442
# The bootloader code continues from sector 1.
$ sudo dd if=bl1.bin.hardkernel of=$DEV conv=fsync bs=512 skip=1 seek=1

# put the at sector 97 (0xC200 = 97*512)
$ sudo dd if=u-boot.gxbb of=$DEV conv=fsync bs=512 seek=97
```

Beware: when defining the DEV variable above, make sure to select the actual microSD card (e.g. **/dev/sda**), and not a potential first partition (e.g. **/dev/sda1**)!

microSD layout

- SD Card
 - see also
 - https://wiki.odroid.com/odroid-c2/software/partition_table#tab__odroid-c21
 - <https://u-boot.readthedocs.io/en/v2021.04/board/amlogic/odroid-c2.html>
 - layout
 - Sector 0: MBR
 - see https://en.wikipedia.org/wiki/Master_boot_record
 - 0x000:
 - usually 440 byte for bootloader code
 - at 0x0DA Windows MBRs have a 6 byte signature
 - 0x0DA: 0x0000
 - 0x0DC: physical drive (0x80–0xFF)
 - 0x0DD: 3 byte timestamp ss:mm:hh (0-59:0-59:0-23)
 - 0x1B8: 4 byte disk signature (since Win2k)
 - 0x1BC: 0x0000 (since Win2k)
 - 0x1BE: 16 byte describing partition #1
 - 0x1CE: 16 byte describing partition #2
 - 0x1DE: 16 byte describing partition #3
 - 0x1EE: 16 byte describing partition #4
 - 0x1FE: 0x55
 - 0x1FF: 0xAA
 - Sector 1 - 96: reserved for BL1 (48.0 KiB)
 - Sector 97 - ... : custom boot loader
 - visible partition start afterwards

Custom card layout:

Area Name	Size	From (sector #)	To (sector #)	Name for fastboot	Partition Name
MBR		0	1		
BL1	48KB	1	96	-	
BL2, BL30, BL301, BL31, U-Boot		97	???	bootloader	
FAT32 for /boot		2048	end	-	/dev/sda1
EXT4 for /rootfs		...	remaining blocks	-	/dev/sda2

Custom Setup

The currently used Buildroot version lacks a U-Boot variant that includes Ethernet support for the ODROID-C2, and thus prevents from utilizing PXE boot. Therefore, a prebuilt U-Boot version from an alternative source (here: Armbian) shall be used instead, and has to be prepared with the **fip_create** tool. As a consequence, the Buildroot build process has to be adapted slightly.

as well as all required binaries (here: from [LibreELEC](#)) shall be used instead. As a consequence, the Buildroot build process has to be adapted slightly.

1. Download the required firmware binaries and tools from [LibreELEC](#)
2. Download a recent Linux image for the ODROID-C2 from the [Armbian](#) website

3. Mount the image and the extract the **u-boot.bin** file (e.g. from `../armbi_root/usr/lib/linux-u-boot-<..>/`)
4. The extracted **u-boot.bin** is equivalent to the **fip.bin** that is generated in the description above; it nevertheless lacks some of the steps required to turn the file into the required **u-boot.gxbb** file
5. Follow the steps described above that are required to convert the **u-boot.bin** (as a replacement of the **fip.bin**) into the **u-boot.gxbb**
6. Adapt the Buildroot configuration for creating the ODROID-C2 SD card image accordingly
 - a. this affects the `../board/hardkernel/odroidc2/genimage.cfg` file
 - b. replace the existing **image** String in the section **partition u-boot** with the name of the U-Boot binary file (here: **u-boot.gxbb**)
7. Run the Buildroot build process and generate a respective SD card image (here: **sdcard.img**)
8. Flash the SD card image to a formatted SD card (e.g. using Balena Etcher or **dd**)
9. Copy the custom **boot.scr** file (which is created as described above) to the boot partition of the SD card and replace the existing file; this then allows for the required PXE boot

Bootflow

TODO: describe boot flow

Binaries

Boot Phase	Software	File	Status	Loaded from	Executed at	Environment	Comment
Phase 1	BootROM code	-	prebuilt	Non-volatile within hard-wired BootROM (iROM)			
Phase 2	BL1	bl1.bin.hardkernel	prebuilt	SD/eMMC			- generated by Amlogic for ODROID-C2 platform - can be redistributed without restriction
	BL30, BL301, BL31, BL33	u-boot.gxbb		SD/eMMC			what is BL30 and BL301 ???

Memory usage during the boot process

TODO: provide memory layout overview

- Boot process
 - DRAM is physically mapped at 0x0
 - U-Boot as BL31 is loaded from SD at 0x00034200 to 0x01000000 (16 MiB)
 - U-Boot loads `/boot/boot.scr` to 0x08000000 (128 MiB)

Bootup

When powering up the board, connect via **picocon** in order to receive respective boot output: from proprietary ODROID bootloader/firmware, from U-Boot and finally from TRENTOS.

```
$ sudo picocom -b 115200 -r -l /dev/ttyUSB0
```

You should then see something like the following output (here: exemplarily running **demo_vm_minimal**, loaded by U-Boot via PXE boot):

```
# GXBB:BL1:08dafd:0a8993;FEAT:EDFC318C;POC:3;RCY:0;EMMC:800;NAND:81;SD:0;READ:0;CHK:0;
TE: 236262
no sdio debug board detected

BL2 Built : 11:44:26, Nov 25 2015.
gxb gfb13a3b-c2 - jcao@wonton

Board ID = 8
set vcck to 1100 mv
set vddee to 1050 mv
CPU clk: 1536MHZ
DDR channel setting: DDR0 Rank0+1 same
DDR0: 2048MB(auto) @ 912MHz(2T)-13
DataBus test pass!
AddrBus test pass!
Load fip header from SD, src: 0x0000c200, des: 0x01400000, size: 0x000000b0
Load bl30 from SD, src: 0x00010200, des: 0x01000000, size: 0x00009f50
Sending bl30.....OK.
```


1.3 MiB/s

```
## Starting application at 0x0795f000 ...
```

```
Jumping to kernel-image entry point...
```

```
load_vm_images@main.c:1083 Loading Kernel: 'linux'
```

```

load_vm_images@main.c:1092 VM: 0x20000000 - 0x21c0a200 (kernel 'linux')
load_vm_images@main.c:1153 Loading DTB: 'linux-dtb'
load_vm_images@main.c:1163 VM: 0x2f000000 - 0x2f002471 (DTB 'linux-dtb')
load_vm_images@main.c:1173 Entering VM at 0x20000000
main_continued@main.c:1477 run VM0 ('vm0')
irq_handler@main.c:738 IRQ 40 Dropped
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xc81004c8, memory fault left
unhandled
on_demand_iterator@main.c:1295 OnDemandInstall: Created device-backed memory for addr 0xc8100000
[ 0.000000] Booting Linux on physical CPU 0x0000000000 [0x410fd034]
[ 0.000000] Linux version 5.7.0 (eckls@DEV-NB-10006) (gcc version 11.3.0 (Buildroot 2022.11.2-72-gc18b427bc7-
dirty), GNU ld (GNU Binutils) 2.38) #1 SMP PREEMPT Fri Apr 21 17:54:31 CEST 2023
[ 0.000000] Machine model: Hardkernel ODROID-C2
[ 0.000000] earlycon: meson0 at MMIO 0x00000000c81004c0 (options '')
[ 0.000000] printk: bootconsole [meson0] enabled
[ 0.000000] efi: UEFI not found.
[ 0.000000] cma: Reserved 32 MiB at 0x000000002d000000
[ 0.000000] NUMA: No NUMA configuration found
[ 0.000000] NUMA: Faking a node at [mem 0x0000000020000000-0x000000002fffffff]
[ 0.000000] NUMA: NODE_DATA [mem 0x2ff76100-0x2ff77fff]
[ 0.000000] Zone ranges:
[ 0.000000]   DMA      [mem 0x0000000020000000-0x000000002fffffff]
[ 0.000000]   DMA32    empty
[ 0.000000]   Normal    empty
[ 0.000000] Movable zone start for each node
[ 0.000000] Early memory node ranges
[ 0.000000]   node    0: [mem 0x0000000020000000-0x000000002fffffff]
[ 0.000000] Initmem setup node 0 [mem 0x0000000020000000-0x000000002fffffff]
[ 0.000000] On node 0 totalpages: 65536
[ 0.000000]   DMA zone: 1024 pages used for memmap
[ 0.000000]   DMA zone: 0 pages reserved
[ 0.000000]   DMA zone: 65536 pages, LIFO batch:15
[ 0.000000] percpu: Embedded 23 pages/cpu s53784 r8192 d32232 u94208
[ 0.000000] pcpu-alloc: s53784 r8192 d32232 u94208 alloc=23*4096
[ 0.000000] pcpu-alloc: [0] 0
[ 0.000000] Detected VIPT I-cache on CPU0
[ 0.000000] CPU features: detected: ARM erratum 845719
[ 0.000000] CPU features: kernel page table isolation forced ON by KASLR
[ 0.000000] CPU features: detected: Kernel page table isolation (KPTI)
[ 0.000000] CPU features: detected: ARM erratum 843419
[ 0.000000] Built 1 zonelists, mobility grouping on. Total pages: 64512
[ 0.000000] Policy zone: DMA
[ 0.000000] Kernel command line: console=ttyAML0,115200n8 root=/dev/mmcblk1p2 rootwait rw nosmp debug
loglevel=8 pci=noms earlycon=meson,0xc81004c0 earlyprintk=serial maxcpus=1
[ 0.000000] Dentry cache hash table entries: 32768 (order: 6, 262144 bytes, linear)
[ 0.000000] Inode-cache hash table entries: 16384 (order: 5, 131072 bytes, linear)
[ 0.000000] mem auto-init: stack:off, heap alloc:off, heap free:off
[ 0.000000] Memory: 182732K/262144K available (13756K kernel code, 2088K rwddata, 7280K rodata, 5504K init,
483K bss, 46644K reserved, 32768K cma-reserved)
[ 0.000000] SLUB: Hwalign=64, Order=0-3, MinObjects=0, CPUs=1, Nodes=1
[ 0.000000] rcu: Preemptible hierarchical RCU implementation.
[ 0.000000] rcu:          RCU event tracing is enabled.
[ 0.000000] rcu:          RCU restricting CPUs from NR_CPUS=256 to nr_cpu_ids=1.
[ 0.000000]          Tasks RCU enabled.
[ 0.000000] rcu: RCU calculated value of scheduler-enlistment delay is 25 jiffies.
[ 0.000000] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=1
[ 0.000000] NR_IRQS: 64, nr_irqs: 64, preallocated irq: 0
[ 0.000000] random: get_random_bytes called from start_kernel+0x310/0x4e4 with crng_init=0
[ 0.000000] arch_timer: cp15 timer(s) running at 24.00MHz (virt).
[ 0.000000] clocksource: arch_sys_counter: mask: 0xffffffffffffff max_cycles: 0x588fe9dc0, max_idle_ns:
440795202592 ns
[ 0.000003] sched_clock: 56 bits at 24MHz, resolution 41ns, wraps every 4398046511097ns
[ 0.008288] Console: colour dummy device 80x25
[ 0.012579] Calibrating delay loop (skipped), value calculated using timer frequency.. 48.00 BogoMIPS
(lpj=96000)
[ 0.022849] pid_max: default: 32768 minimum: 301
[ 0.027545] LSM: Security Framework initializing
[ 0.032164] Mount-cache hash table entries: 512 (order: 0, 4096 bytes, linear)
[ 0.039387] Mountpoint-cache hash table entries: 512 (order: 0, 4096 bytes, linear)
[ 0.048144] rcu: Hierarchical SRCU implementation.
[ 0.052413] EFI services will not be available.

```

```

[ 0.056517] smp: Bringing up secondary CPUs ...
[ 0.061010] smp: Brought up 1 node, 1 CPU
[ 0.065023] SMP: Total of 1 processors activated.
[ 0.069752] CPU features: detected: 32-bit EL0 Support
[ 0.074912] CPU features: detected: CRC32 instructions
[ 0.086953] -----[ cut here ]-----
[ 0.086977] CPU: CPUs started in inconsistent modes
[ 0.086996] WARNING: CPU: 0 PID: 1 at arch/arm64/kernel/smp.c:429 smp_cpus_done+0x6c/0x98
[ 0.099039] Modules linked in:
[ 0.102106] CPU: 0 PID: 1 Comm: swapper/0 Not tainted 5.7.0 #1
[ 0.107964] Hardware name: Hardkernel ODROID-C2 (DT)
[ 0.112953] pstate: 60000005 (nZCv daif -PAN -UA0)
[ 0.117766] pc : smp_cpus_done+0x6c/0x98
[ 0.121703] lr : smp_cpus_done+0x6c/0x98
[ 0.125639] sp : ffff800010013e10
[ 0.128964] x29: ffff800010013e10 x28: 0000000000000000
[ 0.134301] x27: 0000000000000000 x26: 0000000000000000
[ 0.139638] x25: 0000000000000000 x24: 0000000000000000
[ 0.144976] x23: 0000000000000000 x22: 0000000000000000
[ 0.150314] x21: ffff800011760100 x20: ffff800011e0f000
[ 0.155651] x19: ffff800011c26000 x18: ffffffff80000000
[ 0.160989] x17: 0000000000000005 x16: 0000a6c73b451cc2
[ 0.166326] x15: ffff800090013ac7 x14: 0000000000000006
[ 0.171664] x13: ffff800010013ad0 x12: ffff800011c21000
[ 0.177001] x11: ffff800010013e10 x10: ffff800010013e10
[ 0.182339] x9 : ffff800010013e10 x8 : 0000000000000000
[ 0.187676] x7 : 6574726174732073 x6 : ffff800011e1a82f
[ 0.193014] x5 : 0000000000000000 x4 : 0000000000000000
[ 0.198351] x3 : 00000000ffffffff x2 : 0000000000000000
[ 0.203689] x1 : 0000000000000000 x0 : ffff00002f838000
[ 0.209027] Call trace:
[ 0.211478] smp_cpus_done+0x6c/0x98
[ 0.215066] smp_init+0x80/0x90
[ 0.218216] kernel_init_freeable+0xf0/0x244
[ 0.222505] kernel_init+0x14/0x108
[ 0.226005] ret_from_fork+0x10/0x24
[ 0.229594] ---[ end trace 461030b56d30fcae ]---
[ 0.234269] alternatives: patching kernel code
[ 0.239514] devtmpfs: initialized
[ 0.243487] KASLR disabled due to lack of seed
[ 0.246731] clocksource: jiffies: mask: 0xffffffff max_cycles: 0xffffffff, max_idle_ns: 7645041785100000 ns
[ 0.256310] futex hash table entries: 256 (order: 2, 16384 bytes, linear)
[ 0.263923] pinctrl core: initialized pinctrl subsystem
[ 0.269191] thermal_sys: Registered thermal governor 'step_wise'
[ 0.269193] thermal_sys: Registered thermal governor 'power_allocator'
[ 0.274502] DMI not present or invalid.
[ 0.285216] NET: Registered protocol family 16
[ 0.290218] DMA: preallocated 256 KiB pool for atomic allocations
[ 0.295429] audit: initializing netlink subsys (disabled)
[ 0.301583] cpuidle: using governor menu
[ 0.304861] audit: type=2000 audit(0.108:1): state=initialized audit_enabled=0 res=1
[ 0.312655] hw-breakpoint: found 6 breakpoint and 4 watchpoint registers.
[ 0.319405] ASID allocator initialised with 32768 entries
[ 0.325512] Serial: AMBA PL011 UART driver
[ 0.338819] HugeTLB registered 1.00 GiB page size, pre-allocated 0 pages
[ 0.339886] HugeTLB registered 32.0 MiB page size, pre-allocated 0 pages
[ 0.346670] HugeTLB registered 2.00 MiB page size, pre-allocated 0 pages
[ 0.353423] HugeTLB registered 64.0 KiB page size, pre-allocated 0 pages
[ 0.361549] cryptd: max_cpu_qlen set to 1000
[ 0.366943] ACPI: Interpreter disabled.
[ 0.369368] VDDIO_A03V3: supplied by P5V0
[ 0.372892] iommu: Default domain type: Translated
[ 0.377540] vgaarb: loaded
[ 0.380112] SCSI subsystem initialized
[ 0.383785] libata version 3.00 loaded.
[ 0.387729] usbcore: registered new interface driver usbfs
[ 0.393037] usbcore: registered new interface driver hub
[ 0.398375] usbcore: registered new device driver usb
[ 0.403743] pps_core: LinuxPPS API ver. 1 registered
[ 0.408409] pps_core: Software ver. 5.3.6 - Copyright 2005-2007 Rodolfo Giometti <giometti@linux.it>
[ 0.417603] PTP clock support registered

```

```

[ 0.421593] EDAC MC: Ver: 3.0.0
[ 0.425354] FPGA manager framework
[ 0.428169] Advanced Linux Sound Architecture Driver Initialized.
[ 0.434801] clocksource: Switched to clocksource arch_sys_counter
[ 0.440565] VFS: Disk quotas dquot_6.6.0
[ 0.444326] VFS: Dquot-cache hash table entries: 512 (order 0, 4096 bytes)
[ 0.451342] pnp: PnP ACPI: disabled
[ 0.458589] NET: Registered protocol family 2
[ 0.459402] tcp_listen_portaddr_hash hash table entries: 256 (order: 0, 4096 bytes, linear)
[ 0.467488] TCP established hash table entries: 2048 (order: 2, 16384 bytes, linear)
[ 0.475274] TCP bind hash table entries: 2048 (order: 3, 32768 bytes, linear)
[ 0.482454] TCP: Hash tables configured (established 2048 bind 2048)
[ 0.488899] UDP hash table entries: 256 (order: 1, 8192 bytes, linear)
[ 0.495417] UDP-Lite hash table entries: 256 (order: 1, 8192 bytes, linear)
[ 0.502476] NET: Registered protocol family 1
[ 0.507139] RPC: Registered named UNIX socket transport module.
[ 0.512728] RPC: Registered udp transport module.
[ 0.517434] RPC: Registered tcp transport module.
[ 0.522157] RPC: Registered tcp NFSv4.1 backchannel transport module.
[ 0.528638] PCI: CLS 0 bytes, default 64
[ 0.533050] kvm [1]: HYP mode not available
[ 0.538200] Initialise system trusted keyrings
[ 0.541375] workingset: timestamp_bits=44 max_order=16 bucket_order=0
[ 0.552668] squashfs: version 4.0 (2009/01/31) Phillip Lougher
[ 0.554069] NFS: Registering the id_resolver key type
[ 0.558707] Key type id_resolver registered
[ 0.562887] Key type id_legacy registered
[ 0.566882] nfs4filelayout_init: NFSv4 File Layout Driver Registering...
[ 0.573752] 9p: Installing v9fs 9p2000 file system support
[ 0.589172] Key type asymmetric registered
[ 0.589191] Asymmetric key parser 'x509' registered
[ 0.592562] Block layer SCSI generic (bsg) driver version 0.4 loaded (major 245)
[ 0.599979] io scheduler mq-deadline registered
[ 0.604523] io scheduler kyber registered
[ 0.614309] EINJ: ACPI disabled.
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xc883c300, memory fault left
unhandled
on_demand_iterator@main.c:1295 OnDemandInstall: Created device-backed memory for addr 0xc883c000
[ 0.639668] Serial: 8250/16550 driver, 4 ports, IRQ sharing enabled
[ 0.645261] c81004c0.serial: ttyAML0 at MMIO
0xc81004c0 (irq = 6, base_baud = 1500000) is a meson_uart
[ 0.654618] printk: console [ttyAML0] enabled
[ 0.654618] printk: console [ttyAML0] enabled
[ 0.657671] printk: bootconsole [meson0] disabled
[ 0.657671] printk: bootconsole [meson0] disabled
[ 0.667316] msm_serial: driver initialized
[ 0.676175] loop: module loaded
[ 0.676748] megasas: 07.713.01.00-rc1
[ 0.680634] libphy: Fixed MDIO Bus: probed
[ 0.682279] tun: Universal TUN/TAP device driver, 1.6
[ 0.687345] thunder_xcv, ver 1.0
[ 0.690024] thunder_bgx, ver 1.0
[ 0.693256] nicpf, ver 1.0
[ 0.696476] hclge is initializing
[ 0.699192] hns3: Hisilicon Ethernet Network Driver for Hip08 Family - version
[ 0.706306] hns3: Copyright (c) 2017 Huawei Corporation.
[ 0.711604] e1000: Intel(R) PRO/1000 Network Driver - version 7.3.21-k8-NAPI
[ 0.718553] e1000: Copyright (c) 1999-2006 Intel Corporation.
[ 0.724275] e1000e: Intel(R) PRO/1000 Network Driver - 3.2.6-k
[ 0.730023] e1000e: Copyright(c) 1999 - 2015 Intel Corporation.
[ 0.735911] igb: Intel(R) Gigabit Ethernet Network Driver - version 5.6.0-k
[ 0.742788] igb: Copyright (c) 2007-2014 Intel Corporation.
[ 0.748331] igbvf: Intel(R) Gigabit Virtual Function Network Driver - version 2.4.0-k
[ 0.756071] igbvf: Copyright (c) 2009 - 2012 Intel Corporation.
[ 0.762117] sky2: driver version 1.30
[ 0.766041] VFIO - User Level meta-driver version: 0.3
[ 0.771758] ehci_hcd: USB 2.0 'Enhanced' Host Controller (EHCI) Driver
[ 0.777154] ehci-pci: EHCI PCI platform driver
[ 0.781540] ehci-platform: EHCI generic platform driver
[ 0.786760] ehci-orion: EHCI orion driver
[ 0.790710] ehci-exynos: EHCI Exynos driver
[ 0.794873] ohci_hcd: USB 1.1 'Open' Host Controller (OHCI) Driver

```

```

[ 0.800941] ohci-pci: OHCI PCI platform driver
[ 0.805346] ohci-platform: OHCI generic platform driver
[ 0.810558] ohci-exynos: OHCI Exynos driver
[ 0.814928] usbcore: registered new interface driver usb-storage
[ 0.822007] i2c /dev entries driver
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xc88344c0, memory fault left
unhandled
on_demand_iterator@main.c:1295 OnDemandInstall: Created device-backed memory for addr 0xc8834000
[ 0.842983] sdhci: Secure Digital Host Controller Interface driver
[ 0.848965] sdhci: Copyright(c) Pierre Ossman
[ 0.853520] Synopsys Designware Multimedia Card Interface Driver
[ 0.860079] VCC1V8: supplied by VDDIO_A03V3
[ 0.863494] meson-gx-mmc d0074000.mmc: allocated mmc-pwrseq
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xc1104408, memory fault left
unhandled
on_demand_iterator@main.c:1305 OnDemandInstall: Created RAM-backed memory for addr 0xc1104000
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xd0074000, memory fault left
unhandled
on_demand_iterator@main.c:1305 OnDemandInstall: Created RAM-backed memory for addr 0xd0074000
[ 0.932544] sdhci-pltfm: SDHCI platform and OF driver helper
[ 0.933305] ledtrig-cpu: registered to indicate activity on CPUs
[ 0.939397] usbcore: registered new interface driver usbhid
[ 0.944052] usbhid: USB HID core driver
[ 0.950589] NET: Registered protocol family 17
[ 0.952449] 9pnet: Installing 9P2000 support
[ 0.956524] Key type dns_resolver registered
[ 0.960982] registered taskstats version 1
[ 0.964750] Loading compiled-in X.509 certificates
[ 0.980392] TFLASH_VDD: supplied by VDDIO_A03V3
[ 0.982502] meson-gx-mmc d0072000.mmc: Got CD GPIO
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xd0072000, memory fault left
unhandled
on_demand_iterator@main.c:1295 OnDemandInstall: Created device-backed memory for addr 0xd0072000
[ 1.029668] ALSA device list:
[ 1.029689] No soundcards found.
[ 1.036204] Waiting for root device /dev/mmcblk1p2...
[ 1.213245] mmc1: new ultra high speed DDR50 SDHC card at address e624
[ 1.214699] mmcblk1: mmc1:e624 SC16G 14.8 GiB
[ 1.223725] mmcblk1: p1 p2
[ 1.310740] EXT4-fs (mmcblk1p2): recovery complete
[ 1.312296] EXT4-fs (mmcblk1p2): mounted filesystem with ordered data mode. Opts: (null)
[ 1.317983] VFS: Mounted root (ext4 filesystem) on device 179:2.
[ 1.326762] devtmpfs: mounted
[ 1.329126] Freeing unused kernel memory: 5504K
[ 1.331410] Run /sbin/init as init process
[ 1.335358] with arguments:
[ 1.338271] /sbin/init
[ 1.340958] with environment:
[ 1.344062] HOME=/
[ 1.346378] TERM=linux
[ 1.349064] earlyprintk=serial
[ 1.422306] random: fast init done
[ 1.427257] EXT4-fs (mmcblk1p2): re-mounted. Opts: (null)
Starting syslogd: OK
Starting klogd: OK
Running sysctl: OK
Loading needed modules...
[ 1.543102] stmmaceth c9410000.ethernet: IRQ eth_wake_irq not found
[ 1.543723] stmmaceth c9410000.ethernet: IRQ eth_lpi not found
[ 1.549852] stmmaceth c9410000.ethernet: PTP uses main clock
[ 1.555254] stmmaceth c9410000.ethernet: no reset control found
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xc9410020, memory fault left
unhandled
on_demand_iterator@main.c:1295 OnDemandInstall: Created device-backed memory for addr 0xc9410000
[ 1.579859] stmmaceth c9410000.ethernet: User ID: 0x11, Synopsys ID: 0x37
[ 1.586555] stmmaceth c9410000.ethernet: DWMAC1000
vm_memory_handle_fault@guest_memory.c:274 Unable to find reservation for addr: 0xc9411058, memory fault left
unhandled
on_demand_iterator@main.c:1295 OnDemandInstall: Created device-backed memory for addr 0xc9411000
[ 1.610173] stmmaceth c9410000.ethernet: DMA HW capability register supported
[ 1.617220] stmmaceth c9410000.ethernet: RX Checksum Offload Engine supported

```

```
[ 1.624367] stmmaceth c9410000.ethernet: COE Type 2
[ 1.629175] stmmaceth c9410000.ethernet: TX Checksum insertion supported
[ 1.635816] stmmaceth c9410000.ethernet: Wake-Up On Lan supported
[ 1.641879] stmmaceth c9410000.ethernet: Normal descriptors
[ 1.647376] stmmaceth c9410000.ethernet: Ring mode enabled
[ 1.652808] stmmaceth c9410000.ethernet: Enable RX Mitigation via HW Watchdog Timer
[ 1.660401] stmmaceth c9410000.ethernet: device MAC address 00:1e:06:37:9a:ed
[ 1.667804] libphy: stmmac: probed
Initializing random number generator: OK
Saving random seed: [ 1.839750] random: dd: uninitialized urandom read (512 bytes read)
OK
Starting network: OK

Welcome to Buildroot
odroidc2 login:
```

seL4/TRENTOS support

- supported in seL4/CaMkES mainline
- <https://docs.sel4.systems/Hardware/OdroidC2.html> does not contain much help
- platform currently used for the AOS course as UNSW and working there. See <https://github.com/SEL4PROJ/AOS>
- ethernet driver
 - based on U-Boot code
 - mainline platsupport lib at util_libs/libethdrivers seems to be broken, see <https://lists.sel4.systems/hyperkitty/list/devel@sel4.systems/thread/WTa7WNE5SRB4IM2UO6NDKSI4W4IQZYI/>
 - version in AOS branch known to work
- TRENTOS
 - experimental support, some things seem to work
 - good hardware platform for [2020_INT - Virtualization](#)
 - virtualization works, but minimal example still need some memory tweaking

UARTs

- general
 - https://wiki.odroid.com/odroid-c2/application_note/gpio/uart
- all UARTs have the same 5 registers:
 - 0x00: UARTx_WFIFO (write into 64 byte output FIFO)
 - 0x04: UARTx_RFIFO (read from 64 byte input FIFO)
 - 0x08: UARTx_CONTROL
 - 0x0c: UARTx_STATUS
 - 0x10: UARTx_MISC
 - 0x04: UARTx_REG5
- UARTs
 - UART0 ("UART_A"):
 - at 0xc11084c0
 - /soc/bus@c1100000/serial@84c0
 - GPIO: TX=8, RX=10 (TXD1/RXD1)
 - UART1 ("UART_B"):
 - at 0xc11084dc
 - /soc/bus@c1100000/serial@84dc
 - GPIO: TX=3, RX=5 (I2CA_SDA/I2CA_SCL)
 - UART2 ("UART_C"):
 - at 0xc1108700
 - /soc/bus@c1100000/serial@8700
 - GPIO: TX=32, RX=26 (GPIO#224/GPIO#225)
 - UART0-AO ("serial0")
 - at 0xc81004c0
 - /soc/bus@c8100000/serial@4c0
 - 4-pin connector CON5
 - UART1-AO
 - at 0xc81004e0
 - /soc/bus@c8100000/serial@4e0
 - not exposed in a pin?